

FINAL EXPLANATION: CHEMISTRY GRADE 10

Teacher Key: Exemplar Answers and Marking Guide

FINAL EXPLANATION: CHEMISTRY GRADE 10 – SUB-STRAND 2.1: INTRODUCTION TO SALTS – FINAL EXPLANATION

Teacher Key: Exemplar Answers (do not give to students)

INSTRUCTIONS GIVEN TO STUDENTS

This is your Final Explanation for the unit. Use everything you learned in Lessons 1 to 8 to answer the driving question: Why did the fertiliser granules get wet and clump together overnight, the washing soda crystals shrink and form a white crust, but the table salt stay completely the same, and what does understanding salts tell us about how to prepare and use them safely in Kenya? Write your answers in the space under each prompt. Use evidence from the experiments you did, such as the solubility test, the preparation of salts and the watch-glass experiment. Write chemical formulae and balanced equations with state symbols where asked. Write in full sentences and use the scientific words you learned. Link each answer back to the storeroom and laboratory phenomenon.

Part 1: What is a salt, and why does classification not explain the puzzle?

Remember the three substances from Lesson 1: CAN fertiliser (calcium ammonium nitrate, used on maize farms in the Rift Valley), washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) and table salt (NaCl). (a) Define a salt. (b) Name the four types of salts and give one example of each. (c) The fertiliser, washing soda and table salt are all normal salts. Explain why this fact alone cannot explain why they behaved differently overnight.

(a) A salt is a substance formed when a hydrogen atom in an acid is fully or partially replaced by a positive ion. (b) Normal salts: all the hydrogen is replaced, e.g. NaCl or Na_2SO_4 . Acidic salts: the hydrogen is only partly replaced, e.g. NaHCO_3 or NaHSO_4 . Basic salts: contain OH^- or O^{2-} as well as the salt anion, e.g. $\text{Pb}(\text{OH})\text{Cl}$ or $\text{Mg}(\text{OH})\text{Cl}$. Double salts: contain two different cations or two different anions, e.g. alum, $\text{KAl}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$. (c) The CAN fertiliser, washing soda and table salt all belong to the same class, normal salts, yet they behaved differently in the same air. If classification explained the behaviour, they would all have behaved the same. So the explanation must come from other properties of each salt, such as solubility and how it interacts with water in the air, not from the class it belongs to.

Part 2: Solubility and why liquid water is not the whole answer

In Lesson 2 you added seven salts to water in test tubes. (a) Which of the salts you tested dissolved, and which did not? Include the salts from the phenomenon. (b) State the

(a) Dissolved: sodium chloride, potassium nitrate, copper sulphate (blue solution), sodium carbonate and ammonium nitrate (the salt representing CAN). Did not dissolve: calcium carbonate and lead sulphate, which stayed cloudy. (b) All nitrates are soluble. Most chlorides are soluble. Most carbonates are insoluble except those of sodium, potassium and ammonium. Most sulphates are soluble except lead(II) sulphate and barium sulphate. (c) In the test tubes the salts dissolved in liquid water that we added. In the storeroom there was only air and no liquid water. Table salt and washing soda are also soluble, yet they did not behave like the fertiliser, so solubility in liquid water cannot explain the three different results. Something in the air itself, water vapour, must be

solubility rules you used for nitrates, chlorides, carbonates and sulphates. (c) The fertiliser in the storeroom got wet even though no liquid water was added to it. Explain why knowing that a salt is soluble in liquid water is not enough to explain the clumping, and what else must be involved.	interacting with the salts, and each salt must respond to it differently.
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Part 3: How salts are prepared and why the method depends on solubility

(a) Write a balanced equation with state symbols for preparing zinc sulphate from zinc and dilute sulphuric acid, and say how you identified the gas produced. (b) Write a balanced equation with state symbols for preparing calcium chloride from hydrochloric acid and calcium carbonate, and say how you identified the gas. (c) Which method would you use to prepare barium sulphate, and why can you not use the methods in (a) and (b)? Write the net ionic equation. (d) Why do we add the solid in excess when preparing a soluble salt, and how is the excess removed?	(a) $\text{Zn(s)} + \text{H}_2\text{SO}_4\text{(aq)} \rightarrow \text{ZnSO}_4\text{(aq)} + \text{H}_2\text{(g)}$. Hydrogen was identified by the 'pop' test: a burning splint makes a pop sound. (b) $2\text{HCl(aq)} + \text{CaCO}_3\text{(s)} \rightarrow \text{CaCl}_2\text{(aq)} + \text{CO}_2\text{(g)} + \text{H}_2\text{O(l)}$. Carbon dioxide turns limewater milky or cloudy. (c) Barium sulphate is insoluble, so it is made by double decomposition (precipitation). Barium chloride solution and sodium sulphate solution are mixed, and a dense white precipitate forms immediately, which is filtered, washed with distilled water and dried. Net ionic equation: $\text{Ba}^{2+}\text{(aq)} + \text{SO}_4^{2-}\text{(aq)} \rightarrow \text{BaSO}_4\text{(s)}$. Na^+ and Cl^- are spectator ions. The methods in (a) and (b) only work for soluble salts, because the product must stay dissolved in solution so that the leftover solid can be filtered out. An insoluble salt would stay mixed with the excess solid. (d) The solid is added in excess to make sure all the acid is used up, so the salt solution is pure and not contaminated with acid. The unreacted solid is removed by filtration, leaving a clear salt solution.
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Part 4: Explaining the phenomenon using behaviour of salts in air

In Lesson 6 you left calcium chloride, sodium carbonate decahydrate and sodium chloride on watch glasses in classroom air and recorded changes in mass. (a) Describe what happened to each of the three samples and what this	(a) Calcium chloride gained mass and became moist and sticky. Sodium carbonate decahydrate lost mass and developed a white powdery surface. Sodium chloride showed no significant change. (b) A deliquescent salt absorbs water vapour from the air and can dissolve in it. An efflorescent salt loses its water of crystallisation to the air and forms a white powder. A stable salt does not interact significantly with water in the air under normal conditions. (c) CAN fertiliser is deliquescent, or strongly hygroscopic. Its ions attract water molecules from the humid storeroom air, so the granules became wet and sticky and clumped together. Washing soda is efflorescent: the crystals lose water of crystallisation to the air, so they shrank and formed a white powdery crust: $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O} \rightarrow \text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O} + 9\text{H}_2\text{O}$ (the white powder is the monohydrate). Table salt is not affected by water in the air under normal conditions, so it stayed unchanged. All three are soluble salts, but their behaviour in air differs because of how each one
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showed about its mass. (b) Define deliquescent, efflorescent and stable (unaffected) salts. (c) Use these ideas to explain, for each substance, what you saw in the storeroom and laboratory: the CAN fertiliser, the washing soda and the table salt. Write the equation for what happens to washing soda.	interacts with water vapour.
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Part 5: Using salts safely in Kenya	
Use your explanation from Part 4 to answer the second half of the driving question. (a) Give advice to a Rift Valley maize farmer on how to store CAN fertiliser and explain why, in terms of what the salt does in air. (b) Explain how the property of washing soda affects how it must be stored, and why it is still useful in laundry. (c) Choose one other Kenyan use of a salt from Lesson 7 (for example fish preservation at Dunga Beach, ORS, glass making or paint) and explain how the property of the salt makes it suitable. (d) Explain why choosing the right preparation method matters when making salts for use.	(a) The farmer should keep CAN in sealed, airtight bags in a dry place. CAN is deliquescent, so in open air it absorbs water vapour, clumps and becomes sticky, which makes it hard to spread and reduces its effectiveness. (b) Washing soda is efflorescent and loses water of crystallisation to the air, so it must be kept in a closed container to stop it turning into a crust. It is still useful in laundry because it is a soluble salt that works as a water softener by reacting with calcium and magnesium ions. (c) Sodium chloride is rubbed onto tilapia and omena at Dunga Beach, Kisumu. It is soluble and stable in air, so it is a reliable preservative. Alternatively, barium sulphate is used as a paint pigment because it is insoluble and does not dissolve. (d) The method must match the solubility of the salt: soluble salts are made by acid with metal, metal oxide, base or carbonate, and insoluble salts are made by precipitation. Using the correct method gives a pure product that is safe and effective, for example a pure fertiliser for agriculture or a barium meal for medicine.

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Scientific accuracy of salt definitions, classification and solubility rules	All definitions, the four salt types with correct examples and the solubility rules are correct and complete.	Most definitions, types and rules are correct, with minor omissions or one wrong example.	Definitions or rules are missing, incorrect or confused.
Preparation methods and	Balanced equations with correct state	Equations are mostly correct but have	Equations are incorrect or missing, or

chemical equations	symbols, correct gas tests, a correct net ionic equation, and a clear reason why the method depends on solubility.	minor errors in balancing or state symbols, or the reason for the method is only partly explained.	methods are mixed up.
Explanation of the phenomenon (air behaviour)	Correctly explains all three substances using deliquescent, efflorescent and stable behaviour, with evidence from the watch-glass experiment and the washing soda equation.	Explains two or three substances correctly but gives limited evidence or reasoning.	Gives descriptions only, or uses the terms incorrectly.
Link to safe use in Kenya	Gives specific, accurate storage advice and links at least one other Kenyan application to the property of the salt.	Gives correct advice but the link to properties is general.	Advice is vague or not linked to the properties of salts.
Use of evidence and scientific communication	Cites evidence from the lessons, uses scientific vocabulary correctly and writes clearly in full sentences.	Uses some evidence and mostly correct vocabulary.	Little evidence is used and the vocabulary is often incorrect.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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